

(b) Can you prove this starting with the weaker assumption that the upper density $\bar{\mathbf{d}}(A)$ is positive, and reaching the same conclusion?

Exercise 2.2.3. (a) Let $(X, \mathbf{d})$ be a compact metric space and let $T : X \rightarrow X$ be a continuous map. Suppose that μ is a T -invariant probability measure defined on the Borel subsets of X . Prove that for μ -almost every $x \in X$ there is a sequence $n_k \rightarrow \infty$ with $T^{n_k}(x) \rightarrow x$ as $k \rightarrow \infty$.

(b) Prove that the same conclusion holds under the assumption that X is a metric space, $T : X \rightarrow X$ is Borel measurable, and μ is a T -invariant probability measure.

2.3 Ergodicity

Ergodicity is the natural notion of indecomposability in ergodic theory⁽¹⁵⁾. The definition of ergodicity for $(X, \mathcal{B}, \mu, T)$ means that it is impossible to split X into two subsets of positive measure each of which is invariant under T .

Definition 2.13. A measure-preserving transformation $T : X \rightarrow X$ of a probability space $(X, \mathcal{B}, \mu)$ is *ergodic* if for any^{*} $B \in \mathcal{B}$,

$$T^{-1}B = B \implies \mu(B) = 0 \text{ or } \mu(B) = 1. \quad (2.2)$$

When the emphasis is on the map $T : X \rightarrow X$, and we are studying different T -invariant measures, we will also say that μ is an ergodic measure for T . It is useful to have several different characterizations of ergodicity, and these are provided by the following proposition.

Proposition 2.14. *The following are equivalent properties for a measure-preserving transformation T of $(X, \mathcal{B}, \mu)$.*

- (1) T is ergodic.
- (2) For any $B \in \mathcal{B}$, $\mu(T^{-1}B \Delta B) = 0$ implies that $\mu(B) = 0$ or $\mu(B) = 1$.
- (3) For $A \in \mathcal{B}$, $\mu(A) > 0$ implies that $\mu(\bigcup_{n=1}^{\infty} T^{-n}A) = 1$.
- (4) For $A, B \in \mathcal{B}$, $\mu(A)\mu(B) > 0$ implies that there exists $n \geq 1$ with

$$\mu(T^{-n}A \cap B) > 0.$$

- (5) For $f : X \rightarrow \mathbb{C}$ measurable, $f \circ T = f$ almost everywhere implies that f is equal to a constant almost everywhere.

In particular, for an ergodic transformation and countably many sets of positive measure, almost every point visits all of the sets infinitely often under iterations by the ergodic transformation.

^{*} A set $B \in \mathcal{B}$ with $T^{-1}B = B$ is called *strictly invariant* under T .

PROOF OF PROPOSITION 2.14. (1) $\implies$ (2): Assume that T is ergodic, so the implication (2.2) holds, and let B be an *almost invariant* measurable set—that is, a measurable set B with $\mu(T^{-1}B \triangle B) = 0$. We wish to construct an invariant set from B , and this is achieved by means of the following limsup construction. Let

$$C = \bigcap_{N=0}^{\infty} \bigcup_{n=N}^{\infty} T^{-n}B.$$

For any $N \geq 0$,

$$B \triangle \bigcup_{n=N}^{\infty} T^{-n}B \subseteq \bigcup_{n=N}^{\infty} B \triangle T^{-n}B$$

and $\mu(B \triangle T^{-n}B) = 0$ for all $n \geq 1$, since $B \triangle T^{-n}B$ is a subset of

$$\bigcup_{i=0}^{n-1} T^{-i}B \triangle T^{-(i+1)}B,$$

which has zero measure. Let $C_N = \bigcup_{n=N}^{\infty} T^{-n}B$; the sets C_N are nested,

$$C_0 \supseteq C_1 \supseteq \cdots,$$

and $\mu(C_N \triangle B) = 0$ for each N . It follows that $\mu(C \triangle B) = 0$, so

$$\mu(C) = \mu(B).$$

Moreover,

$$T^{-1}C = \bigcap_{N=0}^{\infty} \bigcup_{n=N}^{\infty} T^{-(n+1)}B = \bigcap_{N=0}^{\infty} \bigcup_{n=N+1}^{\infty} T^{-n}B = C.$$

Thus $T^{-1}C = C$, so by ergodicity $\mu(C) = 0$ or 1 , so $\mu(B) = 0$ or 1 .

(2) $\implies$ (3): Let A be a set with $\mu(A) > 0$, and let $B = \bigcup_{n=1}^{\infty} T^{-n}A$. Then $T^{-1}B \subseteq B$; on the other hand $\mu(T^{-1}B) = \mu(B)$ so $\mu(T^{-1}B \triangle B) = 0$. It follows that $\mu(B) = 0$ or 1 ; since $T^{-1}A \subseteq B$ the former is impossible, so $\mu(B) = 1$ as required.

(3) $\implies$ (4): Let A and B be sets of positive measure. By (3),

$$\mu\left(\bigcup_{n=1}^{\infty} T^{-n}A\right) = 1,$$

so

$$0 < \mu(B) = \mu\left(\bigcup_{n=1}^{\infty} B \cap T^{-n}A\right) \leq \sum_{n=1}^{\infty} \mu(B \cap T^{-n}A).$$

It follows that there must be some $n \geq 1$ with $\mu(B \cap T^{-n}A) > 0$.

(4) $\implies$ (1): Let A be a set with $T^{-1}A = A$. Then

$$0 = \mu(A \cap X \setminus A) = \mu(T^{-n}A \cap X \setminus A)$$

for all $n \geq 1$ so, by (4), either $\mu(A) = 0$ or $\mu(X \setminus A) = 0$.

(2) $\implies$ (5): We have seen that if (2) holds, then T is ergodic. Let f be a measurable complex-valued function on X , invariant under T in the stated sense. Since the real and the imaginary parts of f must also be invariant and measurable, we may assume without loss of generality that f is real-valued. Fix $k \in \mathbb{Z}$ and $n \geq 1$ and write

$$A_n^k = \{x \in X \mid f(x) \in [\frac{k}{n}, \frac{k+1}{n})\}.$$

Then $T^{-1}A_n^k \triangle A_n^k \subseteq \{x \in X \mid f \circ T(x) \neq f(x)\}$, a null set, so by (2)

$$\mu(A_n^k) \in \{0, 1\}.$$

For each n , X is the disjoint union $\bigsqcup_{k \in \mathbb{Z}} A_n^k$. It follows that there must be exactly one $k = k(n)$ with $\mu(A_n^{k(n)}) = 1$. Then f is constant on the set

$$Y = \bigcap_{n=1}^{\infty} A_n^{k(n)}$$

and $\mu(Y) = 1$, so f is constant almost everywhere.

(5) $\implies$ (2): If $\mu(T^{-1}B \triangle B) = 0$ then $f = \chi_B$ is a T -invariant measurable function, so by (5) χ_B is a constant almost everywhere. It follows that $\mu(B)$ is either 0 or 1. $\square$

Proposition 2.15. *Bernoulli shifts are ergodic.*

PROOF. Recall the measure-preserving transformation σ defined in Example 2.9 on the measure space $X = \{0, 1, \dots, n\}^{\mathbb{Z}}$ with the product measure μ . Let B denote a σ -invariant measurable set. Then given any $\varepsilon \in (0, 1)$ there is a finite union of cylinder sets A with $\mu(A \triangle B) < \varepsilon$, and hence with $|\mu(A) - \mu(B)| < \varepsilon$. This means A can be described as

$$A = \{x \in X \mid x|_{[-N, N]} \in F\}$$

for some N and some finite set $F \subseteq \{0, 1, \dots, n\}^{[-N, N]}$ (for brevity we write $[a, b]$ for the interval of integers $[a, b] \cap \mathbb{Z}$). It follows that for $M > 2N$,

$$\sigma^{-M}(A) = \{x \in X \mid x|_{[M-N, M+N]} \in F\},$$

where we think of $x|_{[M-N, M+N]}$ as a function on $[-N, N]$ in the natural way, is defined by conditions on a set of coordinates disjoint from $[-N, N]$, so

$$\mu(\sigma^{-M}A \setminus A) = \mu(\sigma^{-M}A \cap X \setminus A) = \mu(\sigma^{-M}A)\mu(X \setminus A) = \mu(A)\mu(X \setminus A). \quad (2.3)$$

Since B is σ -invariant, $\mu(B \triangle \sigma^{-1}B) = 0$. Now

$$\begin{aligned} \mu(\sigma^{-M}A \triangle B) &= \mu(\sigma^{-M}A \triangle \sigma^{-M}B) \\ &= \mu(A \triangle B) < \varepsilon, \end{aligned}$$

so $\mu(\sigma^{-M}A \triangle A) < 2\varepsilon$ and therefore

$$\mu(\sigma^{-M}A \triangle A) = \mu(A \setminus \sigma^{-M}A) + \mu(\sigma^{-M}A \setminus A) < 2\varepsilon. \quad (2.4)$$

Therefore, by (2.3) and (2.4),

$$\begin{aligned} \mu(B)\mu(X \setminus B) &< (\mu(A) + \varepsilon)(\mu(X \setminus A) + \varepsilon) \\ &= \mu(A)\mu(X \setminus A) + \varepsilon\mu(A) + \varepsilon\mu(X \setminus A) + \varepsilon^2 \\ &< \mu(A)\mu(X \setminus A) + 3\varepsilon < 5\varepsilon. \end{aligned}$$

Since ε was arbitrary, this implies that $\mu(B)\mu(X \setminus B) = 0$, so $\mu(B) = 0$ or 1 as required. $\square$

More general versions of this kind of approximation argument appear in Exercises 2.7.3 and 2.7.4.

Proposition 2.16. *The circle rotation $R_\alpha : \mathbb{T} \rightarrow \mathbb{T}$ is ergodic with respect to the Lebesgue measure $m_{\mathbb{T}}$ if and only if α is irrational.*

PROOF. If $\alpha \in \mathbb{Q}$, then we may write $\alpha = \frac{p}{q}$ in lowest terms, so $R_\alpha^q = I_{\mathbb{T}}$ is the identity map. Pick any measurable set $A \subseteq \mathbb{T}$ with $0 < m_{\mathbb{T}}(A) < \frac{1}{q}$. Then

$$B = A \cup R_\alpha A \cup \dots \cup R_\alpha^{q-1}A$$

is a measurable set invariant under R_α with $m_{\mathbb{T}}(B) \in (0, 1)$, showing that R_α is not ergodic.

If $\alpha \notin \mathbb{Q}$ then for any $\varepsilon > 0$ there exist integers m, n, k with $m \neq n$ and $|m\alpha - n\alpha - k| < \varepsilon$. It follows that $\beta = (m - n)\alpha - k$ lies within ε of zero but is not zero, and so the set $\{0, \beta, 2\beta, \dots\}$ considered in $\mathbb{T}$ is ε -dense (that is, every point of $\mathbb{T}$ lies within ε of a point in this set). Thus $(\mathbb{Z}\alpha + \mathbb{Z})/\mathbb{Z} \subseteq \mathbb{T}$ is dense.

Now suppose that $B \subseteq \mathbb{T}$ is invariant under R_α . Then for any $\varepsilon > 0$ choose a function $f \in C(\mathbb{T})$ with $\|f - \chi_B\|_1 < \varepsilon$. By invariance of B we have

$$\|f \circ R_\alpha^n - f\|_1 < 2\varepsilon$$

for all n . Since f is continuous, it follows that

$$\|f \circ R_t - f\|_1 \leq 2\varepsilon$$

for all $t \in \mathbb{R}$. Thus, since $m_{\mathbb{T}}$ is rotation-invariant,

$$\begin{aligned} \left\| f - \int f(t) \, dt \right\|_1 &= \int \left| \int (f(x) - f(x+t)) \, dt \right| dx \\ &\leq \iint |f(x) - f(x+t)| \, dx \, dt \leq 2\varepsilon \end{aligned}$$

by Fubini's theorem (see Theorem A.13) and the triangle inequality for integrals. Therefore

$$\|\chi_B - \mu(B)\|_1 \leq \|\chi_B - f\|_1 + \left\| f - \int f(t) \, dt \right\|_1 + \left\| \int f(t) \, dt - \mu(B) \right\|_1 < 4\varepsilon.$$

Since this holds for every $\varepsilon > 0$ we deduce that χ_B is constant and therefore $\mu(B) \in \{0, 1\}$. Thus for irrational α the transformation R_α is ergodic with respect to Lebesgue measure. $\square$

Proposition 2.17. *The circle-doubling map $T_2 : \mathbb{T} \rightarrow \mathbb{T}$ from Example 2.4 is ergodic (with respect to Lebesgue measure).*

PROOF. By Example 2.8, T_2 and the Bernoulli shift σ on $X = \{0, 1\}^{\mathbb{N}}$ together with the fair coin-toss measure are measurably isomorphic. By Proposition 2.15 the latter is ergodic, and it is clear that measurably isomorphic systems are either both ergodic or both not ergodic. $\square$

Ergodicity (indecomposability in the sense of measure theory) is a universal property of measure-preserving transformations in the sense that every measure-preserving transformation decomposes into ergodic components. This will be shown in Sects. 4.2 and 6.1. In contrast the natural notion of indecomposability in topological dynamics—minimality—does not permit an analogous decomposition (see Exercise 4.2.3).

In Sect. 2.1 we pointed out that in order to check whether a map is measure-preserving it is enough to check this property on a family of sets that generates the σ -algebra. This is not the case when Definition 2.13 is used to establish ergodicity (see Exercise 2.3.2). Using a different characterization of ergodicity does allow this, as described in Exercise 2.7.3(3).

Exercises for Sect. 2.3

Exercise 2.3.1. Show that ergodicity is not preserved under direct products as follows. Find a pair of ergodic measure-preserving systems $(X, \mathcal{B}_X, \mu, T)$ and $(Y, \mathcal{B}_Y, \nu, S)$ for which $T \times S$ is not ergodic with respect to the product measure $\mu \times \nu$.